

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 21 — Organic Synthesis

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	21.1 Organic Synthesis · Multi-Step Reaction Pathways · Functional Group Interconversions (FGI) · Carbon
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	282 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

- 21.1 Functional Group Interconversions (FGI)
- 21.1 Carbon Skeleton Modification
- 21.1 Multi-Step Strategy & Green Chemistry

- Systematic conversion between alkanes, alkenes, halogenoalkanes, alcohols, aldehydes, ketones,
- Carbon chain extension via nucleophilic substitution of halogenoalkanes with cyanide (CN⁻) and
- Retrosynthetic analysis of target molecules; deducing structures of synthetic intermediates; calcul

SUBTOPIC 21.1 — ORGANIC SYNTHESIS (PATHWAYS, STRATEGIES & GREEN CHEMISTRY)

1 Comprehensive Functional Group Interconversions — 9701/22/M/J/23/Q9(a)–(e)

HARD · Syllabus 21.1

Organic synthesis relies on strategic functional group interconversions (FGIs) to convert accessible starting materials into target compounds. Fig. 1.1 displays the interconversion roadmap linking the major AS functional groups.

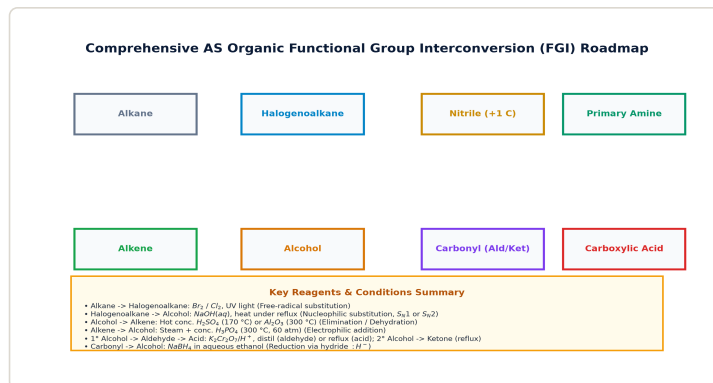


Fig. 1.1: Master functional group interconversion (FGI) roadmap for AS organic chemistry.

((a)) State the reagents and conditions to convert an alkene into:

[3]

- A halogenoalkane
- A primary or secondary alcohol
- An addition polymer

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((b)) State the reagents and conditions to convert a halogenoalkane into:

[3]

- An alcohol
- An alkene
- A nitrile

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((c)) State the reagents and conditions to convert a primary alcohol into:

[3]

- An aldehyde
- A carboxylic acid
- A chloroalkane

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((d)) State the reagent and solvent required to reduce a carboxylic acid to a primary alcohol, and state why NaBH_4 cannot be used for this reduction.

[2]

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2 Three-Step Synthesis: Alkene to Carboxylic Acid — 9701/21/O/N/23/Q4(a)–(c)

HARD · Syllabus 21.1

Propene is converted into propanoic acid in a three-step synthetic sequence.

- ((a))** Step 1: Hydrate propene to an alcohol. Explain why direct hydration with steam/ H_3PO_4 gives predominantly propan-2-ol rather than propan-1-ol. **[2]**

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- ((b))** Devise an alternative two-step route from propene to produce propan-1-ol via an anti-Markovnikov halogenation/substitution sequence. **[3]**

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- ((c))** Describe the reagents and conditions to oxidise propan-1-ol to propanoic acid in high yield. **[2]**

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3 Synthesis of Esters from Alkanes — 9701/22/F/M/22/Q5(a)–(d)

HARD · Syllabus 21.1

Devise a laboratory synthetic route to convert ethane into ethyl ethanoate.

- ((a))** Step 1: Convert ethane into bromoethane. State the reagents and conditions, and identify the type of mechanism. **[3]**

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- ((b))** Step 2: Convert a sample of bromoethane into ethanol. State the reagents and conditions. **[2]**

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- ((c))** Step 3: Convert a second sample of bromoethane into ethanoic acid in two stages via a nitrile. **[3]**

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- ((d))** Step 4: Combine the ethanol and ethanoic acid to produce ethyl ethanoate. Write the equation and state the catalyst and conditions. **[2]**

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4 Synthesis of Amines: Route Selection and Yield — 9701/23/M/J/23/Q4(a)–(c)

HARD · Syllabus 21.1

A chemist considers two synthetic routes to prepare propylamine from 1-chloropropane:

Route A: 1-chloropropane + concentrated ethanolic ammonia in a sealed tube.

Route B: 1-chloropropane → propene → propan-2-ol → propan-2-amine.

((a)) [2]

Identify the major limitation of Route A regarding product purity.

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((b)) [2]

Explain why Route B produces propan-2-amine rather than propylamine.

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((c)) [3]

Devise an optimal two-step route to synthesise pure propylamine from bromoethane with no higher amine contaminants.

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5 Multi-Step Synthesis: Halogenoalkane to 2-Hydroxy Acid — 9701/21/M/J/22/Q5

HARD · Syllabus 21.1

Design a three-step synthetic pathway to prepare 2-hydroxybutanoic acid starting from 1-bromopropane.

((a)) [2]

Step 1: Convert 1-bromopropane into an alkene. State the reagents and conditions.

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((b)) [3]

Step 2: Convert the alkene into an aldehyde. State the intermediate alcohol formed and the oxidising conditions required.

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((c)) [3]

Step 3: Convert the aldehyde into 2-hydroxybutanoic acid via a 2-hydroxynitrile intermediate. State the reagents and conditions for both steps.

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6 Multiple Choice: Optimum Synthetic Step — 9701/12/M/J/23/Q34

EASY · Syllabus 21.1

Which reaction sequence successfully converts but-2-ene into butan-2-one?

((a))

Select the correct sequence:

A Step 1: H_2O / H_3PO_4 (steam); Step 2: Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ under reflux

B Step 1: HBr ; Step 2: KCN ; Step 3: LiAlH_4

C Step 1: H_2 / Ni ; Step 2: Acidified KMnO_4

D Step 1: Aqueous NaOH ; Step 2: Tollens' reagent

A

B

C

D

[1]

((b))

Explain why the product of Step 2 in option A is butan-2-one rather than butanoic acid.

[2]

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Regioselectivity and Isomer Control — 9701/22/O/N/23/Q7

HARD · Syllabus 21.1

When choosing a synthetic route, chemists must minimise the formation of unwanted regioisomers and stereoisomers.

((a))

Explain why adding hydrogen bromide to 2-methylbut-2-ene yields predominantly 2-bromo-2-methylbutane rather than 2-bromo-3-methylbutane.

[3]

((b))

Explain why eliminating HBr from 2-bromobutane with hot ethanolic KOH produces a mixture of three isomeric alkenes. Name all three alkenes.

[3]

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Protecting Groups in Polyfunctional Synthesis — 9701/21/O/N/22/Q6

HARD · Syllabus 21.1

When synthesising a target molecule containing multiple reactive functional groups, chemists often use protecting groups.

((a))

A compound contains both an aldehyde group ($-\text{CHO}$) and a primary alcohol group ($-\text{CH}_2\text{OH}$). A chemist wishes to oxidise ONLY the primary alcohol group to a carboxylic acid while keeping the aldehyde intact. Explain why direct oxidation with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ fails.

[2]

((b))

Explain how converting the aldehyde into an acetal (using ethane-1,2-diol with acid catalyst) protects it during oxidation.

[2]

((c)) State how the original aldehyde is regenerated after the alcohol has been oxidised. [1]

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9 Multi-Step Synthesis: Halogenoalkane to Dicarboxylic Acid — 9701/23/O/N/23/Q5

HARD · Syllabus 21.1

Devise a two-step synthetic pathway to convert 1,2-dibromoethane into butanedioic acid (succinic acid).

((a)) Step 1: State the reagents, conditions, and write the balanced chemical equation for the conversion of 1,2-dibromoethane into a dinitrile. [3]

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((b)) Step 2: State the reagents, conditions, and write the equation for the hydrolysis of the dinitrile to butanedioic acid. [3]

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((c)) State the change in the number of carbon atoms in the main chain from reactant to product. [1]

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10 Multiple Choice: Reagent Differentiation — 9701/11/O/N/23/Q29

EASY · Syllabus 21.1

Which reagent will convert propan-1-ol into 1-bromopropane in a single laboratory step?

((a)) Choose the correct reagent: [1]

- A Aqueous sodium bromide at room temperature
 - B Sodium bromide and concentrated sulfuric acid, heated under reflux
 - C Bromine water at room temperature
 - D Liquid bromine under ultraviolet light
- A
- B
- C
- D

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((b)) Write the equation for the reaction that occurs. [1]

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11 Synthesis of Branched Organic Structures — 9701/21/M/J/21/Q6

HARD · Syllabus 21.1

Devise a multi-step synthetic pathway to prepare 2-methylpropanoic acid from propan-2-ol.

((a)) Step 1: Convert propan-2-ol into 2-bromopropane. State the reagents and conditions. [2]

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((b)) Step 2: Convert 2-bromopropane into 2-methylpropanenitrile. State the reagents and conditions. [2]

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((c)) Step 3: Hydrolyse 2-methylpropanenitrile to 2-methylpropanoic acid. State the reagents and conditions. [2]

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12 Selective Reduction of Carbonyl vs Ester Groups — 9701/22/F/M/21/Q6

HARD · Syllabus 21.1

Compound T contains both a ketone group and an ester group: $\text{CH}_3\text{-CO-CH}_2\text{-COOCH}_2\text{CH}_3$ (ethyl 3-oxobutanoate).

((a)) A chemist wishes to reduce ONLY the ketone group to a secondary alcohol group while leaving the ester group completely unreacted. State the reducing agent and solvent to achieve this selective transformation. [2]

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((b)) Draw the structural formula of the product formed in (a). [1]

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((c)) State what product would be formed if compound T were instead treated with an excess of LiAlH_4 in dry ether. [2]

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13 Separation and Purification in Organic Preparations — 9701/23/O/N/22/Q7

EASY · Syllabus 21.1

During the preparation of a liquid ester in the laboratory, the crude product is separated and purified using four sequential techniques.

((a)) Identify the piece of glassware used to separate the organic layer from the aqueous layer. [1]

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((b)) State the reagent added to the organic layer to remove unreacted carboxylic acid, and state one visible observation. [2]

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((c)) Name a suitable chemical drying agent added to remove dissolved water, and state how you know drying is complete. [2]

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((d)) Name the final purification technique used to obtain the pure, dry ester. [1]

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14 Strategies for Altering Carbon Chain Length — 9701/21/M/J/23/Q10(a)–(d)

HARD · Syllabus 21.1

Modifying carbon framework length is a core challenge in synthesis. Fig. 14.1 compares chain extension (+1 C) via nitriles with chain degradation (−1 C) via the tri-iodomethane reaction.

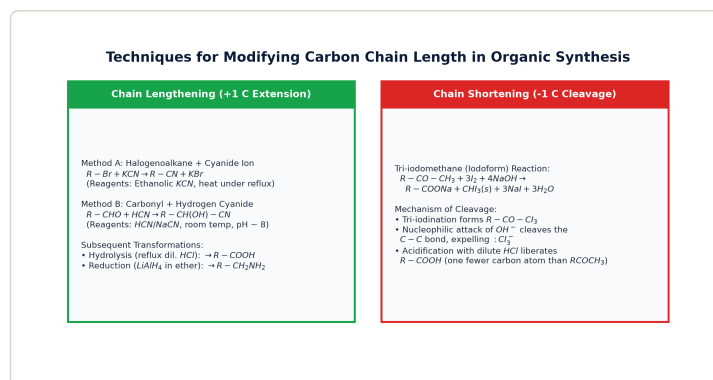


Fig. 14.1: Chemical routes for extending (+1 C) and shortening (−1 C) the carbon skeleton.

- ((a)) Outline two different reaction types that extend the carbon skeleton by exactly one carbon atom, stating the reagents and functional groups involved in each. [4]

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- ((b)) Explain how the tri-iodomethane (iodoform) reaction can be exploited synthetically to shorten a carbon chain by one carbon atom. [3]

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- ((c)) Write the balanced chemical equation for the degradation of butan-2-one with alkaline iodine to form propanoic acid after acidification. [2]

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- ((d)) State the observation that confirms the successful cleavage of the terminal methyl group. [1]

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15 Synthesis of Amino Acids via Strecker Synthesis — 9701/22/O/N/23/Q8

HARD · Syllabus 21.1

The Strecker synthesis is an industrial route to alpha-amino acids starting from aldehydes, ammonia, and hydrogen cyanide.

- ((a)) When ethanal is treated with a mixture of NH_3 and HCN , an alpha-aminonitrile, $CH_3-CH(NH_2)-CN$, is formed. Write the balanced equation for this condensation-addition. [1]

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- ((b))** State the reagents and conditions to hydrolyse $\text{CH}_3\text{-CH}(\text{NH}_2)\text{-CN}$ into 2-aminopropanoic acid (alanine). **[2]**

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- ((c))** Explain why the alanine produced by this chemical synthesis is an optically inactive racemic mixture, unlike natural L-alanine produced by living organisms. **[3]**

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16 Degradation of Alkenes: Ozonolysis vs Permanganate Cleavage — 9701/21/O/N/22/Q7

HARD · Syllabus 21.1

Cleavage of carbon-carbon double bonds can be used to shorten carbon chains or locate double bonds.

- ((a))** When 2-methylbut-2-ene is heated with hot concentrated acidified potassium manganate(VII), the C=C bond is cleaved completely. Deduce the two organic products formed. **[2]**

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- ((b))** State the visible observation during this oxidative cleavage. **[1]**

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- ((c))** Explain why an alkene with a terminal $=\text{CH}_2$ group (such as but-1-ene) produces carbon dioxide gas upon hot acidified KMnO_4 oxidation. **[2]**

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17 Multiple Choice: Step-Up vs Step-Down Reactions — 9701/12/M/J/22/Q27

EASY · Syllabus 21.1

Which reaction results in a decrease in the number of carbon atoms in the main organic molecule?

- ((a))** Select the step-down reaction: **[1]**

- A Bromoethane + KCN in ethanol
- B Ethanal + HCN / NaCN
- C Propanone + alkaline aqueous iodine
- D Ethene + steam / H_3PO_4
- A
- B
- C
- D

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- ((b))** Name the product that contains the eliminated carbon atom. **[1]**

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18 Synthesis of Branched Dicarboxylic Acids — 9701/23/M/J/22/Q5

HARD · Syllabus 21.1

Devise a three-step synthetic pathway to convert 2-methylpropane into 2-methylpropanedioic acid.

((a)) [3]

Step 1: Brominate 2-methylpropane to form a monobromoalkane. State the reagents and explain why 2-bromo-2-methylpropane is the major product.

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((b)) [2]

Step 2: Eliminate HBr to form 2-methylpropene. State the reagents and conditions.

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((c)) [3]

Step 3: React with bromine to form a dibromoalkane, followed by substitution with KCN and acid hydrolysis.

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19 Decarboxylation Reactions (Thermal Chain Shortening) — 9701/21/O/N/23/Q9

HARD · Syllabus 21.1

Heating the anhydrous sodium salt of a carboxylic acid with solid soda lime (NaOH / CaO) results in decarboxylation, removing a carbon atom as carbonate.

((a)) [1]

Write the balanced chemical equation for the decarboxylation of sodium ethanoate, CH₃COONa, with solid sodium hydroxide.

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((b)) [1]

Name the hydrocarbon gas produced.

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((c)) [2]

Explain why soda lime is preferred over pure solid sodium hydroxide in laboratory apparatus.

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20 Grignard Reactions (Extension Carbon-Carbon Coupling) — 9701/22/M/J/23/Q11

HARD · Syllabus 21.1

Organomagnesium reagents (Grignard reagents, RMgX) are widely used to construct carbon-carbon bonds by nucleophilic attack on carbonyl groups.

((a)) [2]

Ethylmagnesium bromide, CH₃CH₂MgBr, is prepared from bromoethane and magnesium. State the solvent required and explain why anhydrous conditions are mandatory.

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((b)) When $\text{CH}_3\text{CH}_2\text{MgBr}$ reacts with methanal followed by dilute acid, propan-1-ol is formed. Write the equation for this transformation. **[1]**

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((c)) Predict whether reaction with propanone produces a primary, secondary, or tertiary alcohol. Name the product. **[2]**

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21**Multi-Step Synthesis: Alkene to Branched Hydroxy Ester — 9701/21/M/J/22/Q9****HARD · Syllabus 21.1**

Devise a synthesis to convert ethene into ethyl 2-hydroxypropanoate (ethyl lactate).

((a)) Outline how ethene can be converted into ethanal in two steps. **[3]**

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((b)) Outline how ethanal is converted into 2-hydroxypropanoic acid via a hydroxynitrile. **[3]**

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((c)) Esterify 2-hydroxypropanoic acid with ethanol, stating the catalyst and conditions. **[2]**

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22**Multiple Choice: Synthetic Pathway to a Diol — 9701/11/M/J/23/Q33****EASY · Syllabus 21.1**

Which reaction converts ethene directly into ethane-1,2-diol in a single step?

((a)) Select the correct reaction: **[1]**

- A Warm with concentrated sulfuric acid
- B Cold, dilute, acidified potassium manganate(VII)
- C Hot, concentrated, acidified potassium manganate(VII)
- D Aqueous sodium hydroxide under reflux
- A
- B
- C
- D

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((b)) State the visible colour change observed during this reaction. **[1]**

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23 Synthesis of Polyfunctional Targets: Paracetamol — 9701/22/F/M/23/Q7**HARD · Syllabus 21.1**

Paracetamol (N-(4-hydroxyphenyl)ethanamide) is synthesised from 4-aminophenol.

((a)) [1]
Draw the structural formula of 4-aminophenol.

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((b)) [2]
4-aminophenol is reacted with ethanoic anhydride, (CH₃CO)₂O. Explain why acylation occurs preferentially at the amino group (-NH₂) rather than at the phenolic hydroxyl group (-OH).

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((c)) [1]
Draw the structural formula of paracetamol.

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24 Isomer Separation and Enantiopurity in Synthesis — 9701/23/O/N/23/Q8**HARD · Syllabus 21.1**

The drug thalidomide was administered as a racemic mixture; the (R)-enantiomer is an effective sedative, whereas the (S)-enantiomer is teratogenic.

((a)) [1]
Define the term racemic mixture.

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((b)) [2]
Explain why standard laboratory syntheses from achiral starting materials invariably produce racemic mixtures rather than a single enantiomer.

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((c)) [2]
State two modern strategies pharmaceutical companies use to obtain enantiopure single-isomer drugs.

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25 Synthesis of Cyclic Ethers and Epoxides — 9701/21/O/N/22/Q9**HARD · Syllabus 21.1**

Epoxides are versatile 3-membered cyclic ethers prepared by treating alkenes with peroxycarboxylic acids or via halohydrins.

((a)) [2]
When ethene reacts with aqueous chlorine, 2-chloroethanol, CH₂Cl-CH₂OH, is formed. Explain why 2-chloroethanol is formed rather than 1,2-dichloroethane when chlorine water is used.

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((b)) [1]
When 2-chloroethanol is treated with cold aqueous sodium hydroxide, an intramolecular S_N2 substitution occurs. Draw the structure of the cyclic product formed.

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((c)) Name this cyclic product.

[1]

26 Multiple Choice: Optimum Catalyst for Hydrogenation — 9701/12/O/N/23/Q29

EASY · Syllabus 21.1

Which catalyst is used for the catalytic addition of hydrogen gas across alkene double bonds?

- ((a)) Select the correct catalyst:
- A Al_2O_3 at 300°C
 - B Nickel (Ni) heated or Platinum (Pt) at room temperature
 - C Concentrated H_2SO_4 at 170°C
 - D FeCl_3 at room temperature
- A
B
C
D

[1]

((b)) State the industrial importance of this reaction in the food industry.

[1]

27 Retrosynthetic Analysis of Target Molecules — 9701/21/M/J/23/Q11(a)–(d)

HARD · Syllabus 21.1

Retrosynthesis involves working backwards from the desired target molecule by identifying strategic disconnections and functional group precursors. Fig. 27.1 outlines the retrosynthetic planning for ethyl propanoate from bromoethane.

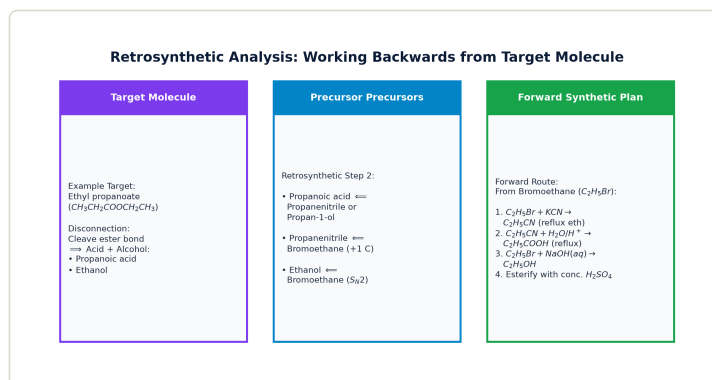


Fig. 27.1: Retrosynthetic disconnection approach to synthesise ethyl propanoate from bromoethane.

((a)) In retrosynthetic analysis, explain what is meant by a 'disconnection' and identify the bond disconnected in ethyl propanoate.

[2]

((b)) State the two precursor molecules generated by disconnecting this bond.

[2]

- ((c)) Show how both precursor molecules can be synthesised independently from bromoethane. [4]

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- ((d)) Write the balanced chemical equation for the final coupling step, stating the catalyst and conditions. [2]

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28 Retrosynthetic Planning for 2-Methylbutanoic Acid — 9701/22/O/N/23/Q9

HARD · Syllabus 21.1

A target molecule is 2-methylbutanoic acid, $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-COOH}$.

- ((a)) Identify a suitable halogenoalkane starting material that can be converted into 2-methylbutanoic acid in a single two-stage sequence. [1]

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- ((b)) Write the structural formula of the nitrile intermediate formed. [1]

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- ((c)) State the reagents and conditions for both steps in the forward synthesis. [3]

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29 Synthesis of N-Phenylethanamide (Acetanilide) — 9701/21/O/N/22/Q8

EASY · Syllabus 21.1

Acetanilide, $\text{C}_6\text{H}_5\text{NHCOCH}_3$, is prepared by acylating phenylamine.

- ((a)) Give two alternative acylating agents that can react with phenylamine to produce acetanilide. [2]

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- ((b)) Write the balanced chemical equation using ethanoyl chloride, stating any by-products. [2]

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- ((c)) Acetanilide is recrystallised from boiling water. Explain how recrystallisation purifies a solid organic product. [2]

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30 Yield Degradation across Multi-Step Routes — 9701/22/M/J/22/Q8

HARD · Syllabus 21.1

A synthetic chemist compares two alternative routes to synthesise target compound X:

- Route 1: A 2-step route where each step has an 80% yield.
- Route 2: A 5-step route where each step has an 85% yield.

((a)) [2]

Calculate the overall percentage yield of Route 1.

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((b)) [2]

Calculate the overall percentage yield of Route 2.

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((c)) [1]

State the general conclusion regarding the number of synthetic steps in chemical route design.

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31 Synthesis of 2-Bromopropanoic Acid from Propene — 9701/23/M/J/23/Q5

HARD · Syllabus 21.1

Devise a synthesis to convert propene into 2-bromopropanoic acid.

((a)) [3]

Step 1: Convert propene into propanoic acid. Outline the necessary reactions and reagents.

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((b)) [2]

Step 2: Brominate propanoic acid selectively at the alpha-carbon using red phosphorus and bromine (Hell-Volhard-Zelinsky conditions). State the role of red phosphorus.

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((c)) [1]

Draw the structural formula of 2-bromopropanoic acid and identify its chiral centre.

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32 Multiple Choice: Retrosynthetic Disconnection of Amides — 9701/11/M/J/23/Q34

EASY · Syllabus 21.1

Which pair of compounds reacts directly at room temperature to form N-methylpropanamide?

((a)) [1]

Select the correct pair:

- A Propanoyl chloride and methylamine
- B Propanoic acid and methanol
- C Propanenitrile and methylamine
- D Ethanoic acid and ethylamine

- A
- B
- C
- D

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((b)) Write the equation for this reaction. [1]

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33 Synthesis of a Cyclic Ketone: Cyclohexanone — 9701/21/O/N/23/Q10

EASY · Syllabus 21.1

Outline a laboratory preparation of cyclohexanone from cyclohexanol.

((a)) State the oxidising agent and reaction conditions required. [2]

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((b)) Write the equation for this oxidation using [O]. [1]

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((c)) State how cyclohexanone can be separated from the green inorganic chromium reaction mixture. [2]

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34 Synthesis of 1,3-Diaminopropane — 9701/22/F/M/22/Q7

HARD · Syllabus 21.1

Devise a two-step synthetic pathway to convert 1,3-dichloropropane into propane-1,3-diamine.

((a)) State the reagents, conditions, and write the balanced chemical equation. [3]

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((b)) Explain why an excess of concentrated ethanolic ammonia is critical in this preparation. [2]

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35 Convergence in Synthesis: Convergent vs Linear Routes — 9701/23/O/N/23/Q9

HARD · Syllabus 21.1

In complex organic synthesis, chemists distinguish between linear synthesis ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow \text{Target}$) and convergent synthesis ($A \rightarrow B$; $C \rightarrow D$; then $B + D \rightarrow \text{Target}$).

((a)) Assume a linear synthesis consists of 4 steps, each with an 80% yield. Calculate the overall percentage yield of the target. [2]

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((b)) Assume a convergent synthesis makes fragment B in 2 steps (80% each) and fragment D in 2 steps (80% each), followed by coupling B and D in one step (80%). Calculate the overall yield of the convergent route based on the longest linear path. [2]

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((c)) State two practical advantages of convergent synthesis in industrial chemical plants. [2]

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36 Multiple Choice: Choosing Reagents for Chlorination — 9701/12/M/J/23/Q35

EASY · Syllabus 21.1

Which reagent is the most convenient for converting a carboxylic acid into an acyl chloride on a laboratory scale?

((a)) [1]

Select the reagent:

- A Cl₂ in UV light
- B NaCl and concentrated H₂SO₄
- C Thionyl chloride (SOCl₂)
- D Concentrated hydrochloric acid

- A
- B
- C
- D

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((b)) Explain why the by-products of this reagent simplify purification. [2]

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37 Synthesis of Esters from Alcohols and Carboxylic Acids — 9701/21/M/J/22/Q10

EASY · Syllabus 21.1

Write a detailed experimental protocol to prepare and isolate pure ethyl ethanoate in the laboratory.

((a)) [3]

List the reactants, catalyst, and heating technique used in the initial reaction.

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((b)) Describe the purification steps using a separating funnel, identifying all reagents used to wash the product. [3]

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((c)) Describe the drying and final collection procedure. [2]

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38 Deducing Multi-Step Intermediate Compounds — 9701/22/F/M/21/Q7

HARD · Syllabus 21.1

Consider the synthetic sequence:

But-1-ene → (Step 1: HBr) → Compound A → (Step 2: aqueous NaOH) → Compound B → (Step 3: acidified K₂Cr₂O₇) → Compound C.

- ((a)) Deduce the structural formula and systematic name of Compound A. [2]
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-
- ((b)) Deduce the structural formula and systematic name of Compound B. [2]
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- ((c)) Deduce the structural formula and systematic name of Compound C, and state a test to confirm the presence of its functional group. [3]
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39 **Synthesis of Polyfunctional Molecule: Aspirin — 9701/23/O/N/23/Q10**

HARD · Syllabus 21.1

Salicylic acid (2-hydroxybenzoic acid) is converted into aspirin (2-ethanoyloxybenzoic acid) by reaction with ethanoic anhydride.

- ((a)) Write the balanced equation for the synthesis of aspirin, showing the structural formulas of all reactants and products. [2]
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- ((b)) State the catalyst used and explain why a water bath is used to warm the mixture. [2]
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- ((c)) Describe how pure aspirin crystals are isolated from the reaction mixture. [2]
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40 **Green Chemistry Metrics: Yield vs Atom Economy — 9701/22/M/J/23/Q12(a)–(d)**

HARD · Syllabus 21.1

Modern industrial organic synthesis prioritises Green Chemistry principles. Fig. 40.1 contrasts Percentage Yield with Percentage Atom Economy.

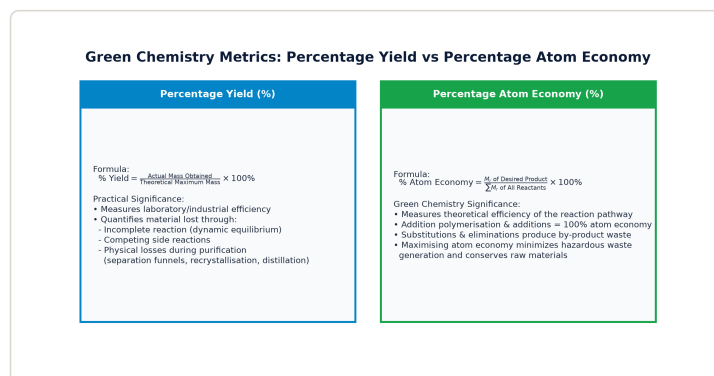


Fig. 40.1: Comparison between percentage yield and percentage atom economy metrics.

((a)) [2]

State the mathematical formulas for:

- (i) Percentage Yield
- (ii) Percentage Atom Economy

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((b)) [3]

In an industrial synthesis of 1-bromobutane from butan-1-ol:



Calculate the percentage atom economy for this reaction (Mr values: butan-1-ol = 74.1, NaBr = 102.9, H₂SO₄ = 98.1, 1-bromobutane = 137.0).

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((c)) [3]

In an experiment, 14.8 g of butan-1-ol yields 19.2 g of 1-bromobutane. Calculate the percentage yield of 1-bromobutane.

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((d)) [2]

Explain why an industrial reaction can have a 100% percentage yield but still generate substantial environmental waste.

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41

Atom Economy Comparison: Substitution vs Addition — 9701/21/O/N/23/Q11

EASY · Syllabus 21.1

Ethanol can be synthesised by two different methods:

Method 1 (Substitution): $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{NaBr}$

Method 2 (Addition): $\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$

((a)) [2]

State the atom economy of Method 2 without performing a calculation. Justify your answer.

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((b)) [2]

Calculate the percentage atom economy of Method 1 (Mr: CH₃CH₂Br = 108.9, NaOH = 40.0, CH₃CH₂OH = 46.0, NaBr = 102.9).

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((c)) [2]

Explain which method is greener from an industrial manufacturing perspective.

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42 Green Chemistry Principles in Organic Synthesis — 9701/22/M/J/22/Q9

HARD · Syllabus 21.1

Paul Anastas and John Warner formulated the 12 Principles of Green Chemistry to guide sustainable chemical production.

((a)) State what is meant by the principle 'Design for Degradation'. **[1]**

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((b)) Explain why catalytic reagents are inherently superior to stoichiometric reagents in sustainable chemical processes. **[2]**

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((c)) State two reasons why water or supercritical CO₂ is preferred over organic solvents like dichloromethane or benzene. **[2]**

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43 Multi-Step Percentage Yield Calculation — 9701/23/O/N/23/Q11

HARD · Syllabus 21.1

A synthesis of propanoic acid from bromoethane involves two steps:

Step 1: $\text{C}_2\text{H}_5\text{Br} + \text{KCN} \rightarrow \text{C}_2\text{H}_5\text{CN} + \text{KBr}$ (yield = 75.0%)

Step 2: $\text{C}_2\text{H}_5\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{COOH} + \text{NH}_4\text{Cl}$ (yield = 80.0%)

((a)) Calculate the overall percentage yield of propanoic acid from bromoethane. **[2]**

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((b)) Calculate the mass of bromoethane ($M_r = 108.9$) required to produce 11.1 g of propanoic acid ($M_r = 74.1$). **[3]**

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44 Multiple Choice: Highest Atom Economy — 9701/12/M/J/23/Q36

EASY · Syllabus 21.1

Which reaction has the highest percentage atom economy for the production of the specified product?

((a)) Select the reaction: **[1]**

- A Polymerisation of ethene to poly(ethene)
- B Hydrolysis of bromoethane to ethanol
- C Oxidation of ethanol to ethanoic acid
- D Esterification of ethanoic acid with ethanol

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((b)) State the percentage atom economy of your selected reaction. **[1]**

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Atom Economy in Ibuprofen Synthesis (Boots vs BHC Route) — 9701/21/M/J/22/Q11

HARD · Syllabus 21.1

The original Boots synthesis of the analgesic ibuprofen required 6 steps and had an atom economy of 40%. The modern BHC catalytic synthesis requires only 3 steps and achieves an atom economy of 77% (or 99% when acetic acid by-product is recovered).

((a)) State two reasons why the BHC route is superior to the Boots route under Green Chemistry principles. **[2]**

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((b)) Explain why achieving higher atom economy reduces the environmental footprint of a pharmaceutical factory. **[2]**

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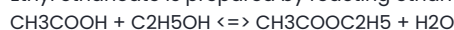
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Calculation of Atom Economy for Esterification — 9701/22/F/M/22/Q8

EASY · Syllabus 21.1

Ethyl ethanoate is prepared by reacting ethanoic acid with ethanol:



((a)) Calculate the percentage atom economy for the production of ethyl ethanoate (Mr: $\text{CH}_3\text{COOH} = 60.0$, $\text{C}_2\text{H}_5\text{OH} = 46.0$, $\text{CH}_3\text{COOC}_2\text{H}_5 = 88.0$, $\text{H}_2\text{O} = 18.0$). **[2]**

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((b)) Calculate the percentage atom economy if the reaction were performed using ethanoyl chloride: $\text{CH}_3\text{COCl} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{HCl}$ (Mr: $\text{CH}_3\text{COCl} = 78.5$, $\text{HCl} = 36.5$). **[2]**

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((c)) Compare the atom economies of the two routes and comment on practical laboratory choice. **[2]**

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47

Catalyst Recyclability in Green Synthesis — 9701/21/O/N/23/Q12

HARD · Syllabus 21.1

Heterogeneous solid catalysts are widely preferred over homogeneous liquid acid catalysts in industrial organic synthesis.

((a)) Explain the difference between a homogeneous catalyst and a heterogeneous catalyst. **[2]**

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((b)) State two reasons why passing alcohol vapour over solid aluminium oxide catalyst granules is greener than heating with concentrated sulfuric acid. **[2]**

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48 Multiple Choice: Atom Economy of Addition Reactions — 9701/11/O/N/23/Q30

EASY · Syllabus 21.1

What is the theoretical atom economy for the electrophilic addition of hydrogen chloride to propene to make 2-chloropropane?

((a))

Select the atom economy:

- A 50%
 - B 75%
 - C 100%
 - D Dependant on the temperature
- A
B
C
D

[1]**((b))**

Explain why all electrophilic addition reactions have this atom economy.

[1]

49 Life Cycle Assessment in Industrial Route Evaluation — 9701/22/M/J/23/Q13

HARD · Syllabus 21.1

When evaluating a synthetic pathway, an industrial chemist assesses the Environmental Factor (E-factor), defined as: $E\text{-factor} = \frac{\text{total mass of waste produced}}{\text{mass of desired product}}$.

((a))

A pharmaceutical process manufactures 25.0 kg of an active ingredient while generating 1,250 kg of chemical waste. Calculate the E-factor for this process.

[2]**((b))**

State whether a greener, more sustainable process has a higher or lower E-factor. Explain your answer.

[2]**((c))**

Identify two major sources of waste in pharmaceutical production that contribute to high E-factors.

[2]

50 Comprehensive Retrosynthetic Design Problem — 9701/21/M/J/23/Q12

HARD · Syllabus 21.1

Devise a complete, practical synthetic pathway to prepare 2-hydroxy-2-methylbutanoic acid starting from but-2-ene.

((a))

Step 1: Convert but-2-ene into butan-2-ol. State the reagents and conditions.

[2]**((b))**

Step 2: Oxidise butan-2-ol to butanone. State the reagents, conditions, and write the equation using [O].

[2]

((c))

Step 3: React butanone with HCN to form a 2-hydroxynitrile. State the reagents, catalyst, and mechanism.

[3]

((d))

Step 4: Hydrolyse the 2-hydroxynitrile to 2-hydroxy-2-methylbutanoic acid. State the reagents and conditions, and explain why the final product is an optically inactive racemate.

[3]

Mark Scheme — Topic 21 — Organic Synthesis

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	(i) Hydrogen halide (e.g. HBr(g) or HCl(g)) at room temperature (electrophilic addition) [1] (ii) Steam (H ₂ O(g)) and concentrated H ₃ PO ₄ catalyst at 300 °C, 60 atm [1] (iii) High pressure (~1500–3000 atm) with trace O ₂ initiator (or Ziegler-Natta catalyst) [1]	3
1(b)	(i) Aqueous sodium hydroxide (NaOH(aq)), heat under reflux (nucleophilic substitution) [1] (ii) Ethanolic potassium hydroxide (KOH in ethanol), heat under reflux (elimination) [1] (iii) Potassium cyanide (KCN) in aqueous ethanol, heat under reflux (nucleophilic substitution) [1]	3
1(c)	(i) Acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄), heat gently with immediate distillation [1] (ii) Acidified potassium dichromate(VI) in excess, heat under reflux [1] (iii) Solid PCl ₅ at room temperature (or SOCl ₂ at room temperature) [1]	3
1(d)	Reagent/solvent: LiAlH ₄ in dry ether (followed by dilute acid workup) [1] The carbonyl carbon of a carboxylic acid is weakly electrophilic due to electron donation from the adjacent –OH oxygen lone pair; NaBH ₄ is too weak a nucleophile to transfer hydride to it [1]	2
2(a)	Electrophilic addition proceeds via the more stable secondary carbocation (CH ₃ –CH–CH ₃) rather than the primary carbocation [1] The secondary carbocation is stabilised by the positive inductive effect (+I) of two electron-releasing methyl groups (Markovnikov's rule) [1]	2
2(b)	Step 1: Add HBr in the presence of organic peroxides (free radical addition) to form 1-bromopropane [2] Step 2: Hydrolyse 1-bromopropane with aqueous NaOH under reflux to yield propan-1-ol [1]	3
2(c)	Reagent: Excess acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄) [1] Condition: Heat under reflux (to ensure complete oxidation through the aldehyde stage to the acid) [1]	2
3(a)	Reagents/conditions: Bromine (Br ₂) and ultraviolet (UV) light / sunlight [1] Mechanism: Free-radical substitution [1] C ₂ H ₆ + Br ₂ → C ₂ H ₅ Br + HBr [1]	3
3(b)	Reagents/conditions: Aqueous sodium hydroxide (NaOH(aq)), heat under reflux (nucleophilic substitution) [2]	2
3(c)	Stage 1: Bromoethane + KCN in aqueous ethanol, heat under reflux → Propanenitrile (+1 C would give propanoic acid, wait: for ethanoic acid, oxidise ethanol directly with excess acidified K ₂ Cr ₂ O ₇ under reflux!) [2] Oxidation of ethanol with excess acidified K ₂ Cr ₂ O ₇ under reflux gives ethanoic acid (2 carbons) [1]	3
3(d)	CH ₃ COOH + CH ₃ CH ₂ OH ⇌ CH ₃ COOCH ₂ CH ₃ + H ₂ O [1] Catalyst/conditions: Concentrated sulfuric acid (conc. H ₂ SO ₄), heat under reflux [1]	2
4(a)	Produces a complex mixture of primary, secondary, and tertiary amines and quaternary ammonium salts due to consecutive nucleophilic substitution [1] Requires fractional distillation to separate, reducing the yield of pure propylamine [1]	2
4(b)	Hydration of propene follows Markovnikov's rule, adding –OH to the secondary carbon to form propan-2-ol [1] Subsequent amination retains substitution at C2, forming branched propan-2-amine rather than linear propylamine [1]	2
4(c)	Step 1: Bromoethane + KCN in aqueous ethanol, heat under reflux → Propanenitrile (CH ₃ CH ₂ CN) [2] Step 2: Reduce propanenitrile with LiAlH ₄ in dry ether (or H ₂ /Ni) → Propylamine (CH ₃ CH ₂ CH ₂ NH ₂) [1]	3
5(a)	Reagents/conditions: Ethanolic potassium hydroxide (KOH in ethanol), heat under reflux (elimination of HBr to propene) [2]	2
5(b)	Hydrate to propan-1-ol via anti-Markovnikov route or convert to primary halogenoalkane then alcohol [1] Oxidise propan-1-ol with acidified K ₂ Cr ₂ O ₇ and immediate distillation to obtain propanal [2]	3
5(c)	React propanal with HCN and NaCN catalyst at room temp / pH 8 → 2-hydroxybutanenitrile [2] Heat 2-hydroxybutanenitrile under reflux with dilute hydrochloric acid → 2-hydroxybutanoic acid [1]	3
6(a)	A (Step 1: H ₂ O / H ₃ PO ₄ (steam); Step 2: Acidified K ₂ Cr ₂ O ₇ under reflux) [1]	1
6(b)	Hydration of but-2-ene yields butan-2-ol, which is a secondary (2°) alcohol [1] Secondary alcohols oxidise to ketones (butan-2-one) and cannot be oxidised further to carboxylic acids under non-destructive conditions [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
7(a)	Electrophilic addition of H can generate either a tertiary carbocation ((CH ₃) ₂ C-CH ₂ CH ₃) or a secondary carbocation ((CH ₃) ₂ CH-CH-CH ₃) [1] The tertiary carbocation has three electron-donating alkyl groups that stabilise the positive charge via the inductive effect [1] Tertiary carbocation forms faster (lower activation energy), yielding predominantly 2-bromo-2-methylbutane [1]	3
7(b)	A proton can be eliminated from either C1 or C3 adjacent to the C-Br bond [1] Elimination from C1 yields but-1-ene; elimination from C3 yields but-2-ene [1] But-2-ene exhibits geometric isomerism, existing as cis-but-2-ene and trans-but-2-ene (total 3 alkenes) [1]	3
8(a)	Aldehydes are more easily oxidised than primary alcohols [1] Any oxidising agent strong enough to oxidise the alcohol will immediately oxidise the aldehyde group to a carboxylic acid as well [1]	2
8(b)	Acetals are diether derivatives that lack carbonyl unsaturation [1] They are completely stable and inert to oxidising agents in basic or neutral conditions, protecting the carbonyl carbon from oxidation [1]	2
8(c)	Hydrolyse with dilute aqueous acid (acid-catalysed deprotection) [1]	1
9(a)	Reagents: Potassium cyanide (KCN) in excess [1] Conditions: Aqueous ethanol, heat under reflux [1] Br-CH ₂ -CH ₂ -Br + 2KCN → NC-CH ₂ -CH ₂ -CN + 2KBr [1]	3
9(b)	Reagents/conditions: Dilute hydrochloric acid (dilute HCl(aq)), heat under reflux [1] NC-CH ₂ -CH ₂ -CN + 4H ₂ O + 2HCl → HOOC-CH ₂ -CH ₂ -COOH + 2NH ₄ Cl [2]	3
9(c)	Increases by 2 carbon atoms (from 2 carbons in 1,2-dibromoethane to 4 carbons in butanedioic acid) [1]	1
10(a)	B (Sodium bromide and concentrated sulfuric acid, heated under reflux) [1]	1
10(b)	CH ₃ CH ₂ CH ₂ OH + HBr → CH ₃ CH ₂ CH ₂ Br + H ₂ O (generated from NaBr + H ₂ SO ₄) [1]	1
11(a)	Reagents/conditions: NaBr and concentrated H ₂ SO ₄ (or PBr ₃), heat under reflux [2]	2
11(b)	Reagents/conditions: KCN (or NaCN) in aqueous ethanol, heat under reflux [2]	2
11(c)	Reagents/conditions: Dilute hydrochloric acid (or dilute H ₂ SO ₄), heat under reflux [2]	2
12(a)	Reagent: Sodium borohydride (NaBH ₄) [1] Solvent: Aqueous ethanol (or methanol) at room temperature [1]	2
12(b)	CH ₃ -CH(OH)-CH ₂ -COOCH ₂ CH ₃ (ethyl 3-hydroxybutanoate) [1]	1
12(c)	LiAlH ₄ reduces BOTH the ketone (to a 2° alcohol) and the ester group (to two alcohols) [1] Produces butane-1,3-diol (HO-CH ₂ -CH ₂ -CH(OH)-CH ₃) and ethanol (CH ₃ CH ₂ OH) [1]	2
13(a)	Separating funnel [1]	1
13(b)	Aqueous sodium carbonate (Na ₂ CO ₃) or sodium hydrogencarbonate (NaHCO ₃) [1] Effervescence / bubbling of CO ₂ gas [1]	2
13(c)	Anhydrous magnesium sulfate (MgSO ₄) or anhydrous calcium chloride (CaCl ₂) [1] The cloudy liquid turns clear / drying agent remains granular and flows freely [1]	2
13(d)	Simple distillation / fractional distillation [1]	1
14(a)	Method 1: Nucleophilic substitution of a halogenoalkane using potassium cyanide (KCN) in aqueous ethanol under reflux (converts R-X to R-CN) [2] Method 2: Nucleophilic addition of hydrogen cyanide (HCN / NaCN at pH 8) to an aldehyde or ketone (converts R-CHO to 2-hydroxynitrile R-CH(OH)-CN) [2]	4
14(b)	Compounds containing the methyl carbonyl (CH ₃ -CO-R) group undergo tri-iodination to CI ₃ -CO-R [1] Hydroxide ion attacks the carbonyl carbon, cleaving the C-C bond and expelling the :CI ₃ leaving group [1] This removes the terminal methyl group as solid CHI ₃ , leaving behind a carboxylate salt with one fewer carbon atom, which yields RCOOH upon acidification [1]	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
14(c)	$\text{CH}_3\text{COCH}_2\text{CH}_3 + 3\text{I}_2 + 4\text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{COONa} + \text{CHI}_3 + 3\text{NaI} + 3\text{H}_2\text{O}$ [1] $\text{CH}_3\text{CH}_2\text{COONa} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{NaCl}$ [1]	2
14(d)	Formation of a pale yellow crystalline precipitate of tri-iodomethane (CHI_3) with a characteristic medicinal / antiseptic odour [1]	1
15(a)	$\text{CH}_3\text{CHO} + \text{NH}_3 + \text{HCN} \rightarrow \text{CH}_3\text{CH}(\text{NH}_2)\text{CN} + \text{H}_2\text{O}$ [1]	1
15(b)	Dilute hydrochloric acid (dilute $\text{HCl}(\text{aq})$), heat under reflux [1] Followed by neutralisation to isoelectric point [1]	2
15(c)	The intermediate imine / aldehyde precursor is planar around the sp^2 carbon atom [1] Nucleophilic cyanide attack has an equal probability from the top face or bottom face [1] This produces an equimolar (50:50) mixture of both D- and L-enantiomers (racemate), whereas biological synthesis uses stereospecific enzyme catalysts that generate exclusively one enantiomer [1]	3
16(a)	Propanone, $(\text{CH}_3)_2\text{C}=\text{O}$ [1] Ethanoic acid, CH_3COOH [1]	2
16(b)	Purple solution turns colourless (decolourises) [1]	1
16(c)	The terminal $=\text{CH}_2$ group is first oxidised to methanoic acid (HCOOH) [1] Methanoic acid contains an aldehyde group and is further oxidised by hot KMnO_4 to carbon dioxide and water: $\text{HCOOH} + [\text{O}] \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ [1]	2
17(a)	C (Propanone + alkaline aqueous iodine) [1]	1
17(b)	Tri-iodomethane (CHI_3) [1]	1
18(a)	Reagents: Bromine (Br_2) and UV light [1] The tertiary radical $(\text{CH}_3)_3\text{C}^\bullet$ is stabilised by hyperconjugation and positive inductive effects of three methyl groups [1] Forms much faster than primary radicals, giving 2-bromo-2-methylpropane as major product [1]	3
18(b)	Ethanol potassium hydroxide (KOH in ethanol), heat under reflux [2]	2
18(c)	Add Br_2 in CCl_4 to form 1,2-dibromo-2-methylpropane [1] Reflux with excess ethanolic KCN to form dinitrile [1] Reflux with dilute hydrochloric acid to hydrolyse dinitrile to dicarboxylic acid [1]	3
19(a)	$\text{CH}_3\text{COONa} + \text{NaOH} \rightarrow \text{CH}_4 + \text{Na}_2\text{CO}_3$ [1]	1
19(b)	Methane [1]	1
19(c)	Pure NaOH is deliquescent, melts at a relatively low temperature, and attacks/corrodes hot laboratory glassware severely [1] Calcium oxide in soda lime keeps the mixture solid and porous, preventing fusion and protecting the glass boiling tube [1]	2
20(a)	Solvent: Dry ether (ethoxyethane) [1] Grignard reagents are powerful bases that react violently with traces of water: $\text{RMgBr} + \text{H}_2\text{O} \rightarrow \text{RH} + \text{Mg}(\text{OH})\text{Br}$ [1]	2
20(b)	$\text{CH}_3\text{CH}_2\text{MgBr} + \text{HCHO} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OMgBr} \xrightarrow{-(\text{H}_3\text{O})} \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ [1]	1
20(c)	Tertiary (3°) alcohol [1] 2-methylbutan-2-ol [1]	2
21(a)	Step 1: Ethene + steam / H_3PO_4 (300 °C, 60 atm) $\rightarrow$ Ethanol [1] Step 2: Ethanol + acidified $\text{K}_2\text{Cr}_2\text{O}_7$, heat gently with immediate distillation $\rightarrow$ Ethanal [2]	3
21(b)	React ethanal with $\text{HCN} + \text{NaCN}$ catalyst at room temp / pH 8 $\rightarrow$ 2-hydroxypropanenitrile [2] Reflux 2-hydroxypropanenitrile with dilute HCl $\rightarrow$ 2-hydroxypropanoic acid [1]	3
21(c)	React with ethanol in presence of concentrated H_2SO_4 catalyst, heat under reflux [2]	2
22(a)	B (Cold, dilute, acidified potassium manganate(VII)) [1]	1
22(b)	Purple solution turns colourless (or forms brown precipitate of MnO_2 if neutral/alkaline) [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
23(a)	Benzene ring with -OH at position 1 and -NH ₂ at position 4 (para) [1]	1
23(b)	Nitrogen is less electronegative than oxygen (electronegativity 3.0 vs 3.5) [1] The lone pair on nitrogen is more nucleophilic and readily donated than the lone pair on the phenolic oxygen, reacting faster with electrophilic carbonyl carbon [1]	2
23(c)	HO-C ₆ H ₄ -NHCOCH ₃ (or displayed formula) [1]	1
24(a)	An equimolar (50:50) mixture of two optical enantiomers with zero net optical rotation [1]	1
24(b)	Reactions involving planar trigonal (sp ²) carbocation intermediates or planar carbonyl groups have equal activation energy for attack from either face [1] Both enantiomers form at exactly equal rates, giving an equimolar mixture [1]	2
24(c)	Use of chiral catalysts / enantioselective asymmetric synthesis [1] Use of biological enzymes / chiral pools / resolution with chiral resolving acids/bases [1]	2
25(a)	Electrophilic attack of Cl creates a chloronium intermediate [1] Water molecules are present in enormous excess compared to chloride ions and attack the cyclic intermediate preferentially as the nucleophile [1]	2
25(b)	Three-membered cyclic ether: Oxirane (ethylene oxide) [1]	1
25(c)	Oxirane (or epoxyethane) [1]	1
26(a)	B (Nickel (Ni) heated or Platinum (Pt) at room temperature) [1]	1
26(b)	Hydrogenation of liquid unsaturated vegetable oils to produce solid/semi-solid margarines and spreads [1]	1
27(a)	An analytical process of hypothetically breaking a bond in the target molecule to generate simpler precursor fragments (synthons) [1] The ester C-O single bond connecting the carbonyl group to the ethyl group (CH ₃ CH ₂ CO-OCH ₂ CH ₃) [1]	2
27(b)	Propanoic acid (CH ₃ CH ₂ COOH) [1] Ethanol (CH ₃ CH ₂ OH) [1]	2
27(c)	Ethanol: Bromoethane + aqueous NaOH, heat under reflux (nucleophilic substitution) [2] Propanoic acid: Bromoethane + KCN in aqueous ethanol, heat under reflux → Propanenitrile; then reflux propanenitrile with dilute HCl → Propanoic acid [2]	4
27(d)	CH ₃ CH ₂ COOH + CH ₃ CH ₂ OH ⇌ CH ₃ CH ₂ COOCH ₂ CH ₃ + H ₂ O [1] Concentrated H ₂ SO ₄ catalyst, heat under reflux [1]	2
28(a)	2-bromobutane, CH ₃ -CH ₂ -CH(Br)-CH ₃ (or 2-chlorobutane) [1]	1
28(b)	CH ₃ -CH ₂ -CH(CH ₃)-CN (2-methylbutanenitrile) [1]	1
28(c)	Step 1: 2-bromobutane + KCN in aqueous ethanol, heat under reflux [2] Step 2: Heat under reflux with dilute hydrochloric acid [1]	3
29(a)	Ethanoyl chloride (CH ₃ COCl) [1] Ethanoic anhydride ((CH ₃ CO) ₂ O) [1]	2
29(b)	C ₆ H ₅ NH ₂ + CH ₃ COCl → C ₆ H ₅ NHCOCH ₃ + HCl [2]	2
29(c)	Dissolve crude product in minimum volume of hot solvent; insoluble impurities filtered hot [1] Upon slow cooling, pure acetanilide crystals precipitate out while soluble impurities remain dissolved in cold mother liquor [1]	2
30(a)	Overall yield = 0.80 x 0.80 = 0.640 = 64.0% [2]	2
30(b)	Overall yield = (0.85) = 0.4437 = 44.4% [2]	2
30(c)	Routes with fewer synthetic steps are inherently superior because cumulative product losses at each stage severely diminish the overall yield [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
31(a)	Hydrate propene to propan-1-ol via anti-Markovnikov HBr addition then NaOH(aq) hydrolysis [2] Oxidise propan-1-ol with excess acidified K ₂ Cr ₂ O ₇ under reflux → Propanoic acid [1]	3
31(b)	Red phosphorus reacts with bromine to form PBr ₃ in situ [1] PBr ₃ converts a small amount of acid to acyl bromide, catalysing enolisation and selective alpha-bromination [1]	2
31(c)	CH ₃ -C*(H)(Br)-COOH with asterisk on C2 [1]	1
32(a)	A (Propanoyl chloride and methylamine) [1]	1
32(b)	CH ₃ CH ₂ COCl + CH ₃ NH ₂ → CH ₃ CH ₂ CONHCH ₃ + HCl [1]	1
33(a)	Acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄) [1] Heat under reflux (or warm gently) [1]	2
33(b)	C ₆ H ₁₁ OH + [O] → C ₆ H ₁₀ O + H ₂ O [1]	1
33(c)	Steam distillation or direct simple distillation of the organic ketone [1] Separate organic layer using a separating funnel, dry with anhydrous MgSO ₄ , and redistil [1]	2
34(a)	Reagent: Excess concentrated ammonia (NH ₃) in ethanol [1] Condition: Heat under pressure in a sealed tube [1] Cl-CH ₂ -CH ₂ -CH ₂ -Cl + 4NH ₃ → H ₂ N-CH ₂ -CH ₂ -CH ₂ -NH ₂ + 2NH ₄ Cl [1]	3
34(b)	Dihaloalkanes are prone to intermolecular cross-linking and intramolecular cyclisation to form cyclic secondary amines (azetidine / pyrrolidine rings) [1] A large excess of NH ₃ ensures nucleophilic substitution by ammonia dominates, preventing polymerisation and cyclisation [1]	2
35(a)	Linear yield = (0.80) = 0.4096 = 41.0% [2]	2
35(b)	Longest linear sequence = 3 steps (two steps to fragment + one coupling step) [1] Convergent yield = (0.80) ³ = 0.512 = 51.2% [1]	2
35(c)	Higher overall yield of final product [1] Fragments B and D can be manufactured in parallel in separate reactors, saving production time and avoiding risking valuable intermediates in long sequential chains [1]	2
36(a)	C (Thionyl chloride (SOCl ₂)) [1]	1
36(b)	Both by-products, SO ₂ and HCl, are gases that bubble out of the reaction mixture [1] Leaves behind the pure liquid acyl chloride without requiring fractional distillation [1]	2
37(a)	Reactants: Glacial ethanoic acid and pure ethanol [1] Catalyst: Concentrated sulfuric acid (conc. H ₂ SO ₄) [1] Heating under reflux with a water bath or heating mantle (prevent naked flames) [1]	3
37(b)	Distil off crude mixture and transfer to separating funnel [1] Shake with aqueous sodium carbonate (Na ₂ CO ₃) to neutralise and remove unreacted acids (release CO ₂ pressure) [1] Wash organic layer with aqueous calcium chloride (CaCl ₂) to remove unreacted ethanol [1]	3
37(c)	Dry the organic layer with anhydrous magnesium sulfate (MgSO ₄) until clear [1] Filter and fractionally distil, collecting the fraction boiling between 75–78 °C (pure ethyl ethanoate) [1]	2
38(a)	Structure: CH ₃ -CH ₂ -CH(Br)-CH ₃ [1] Systematic name: 2-bromobutane (Markovnikov electrophilic addition) [1]	2
38(b)	Structure: CH ₃ -CH ₂ -CH(OH)-CH ₃ [1] Systematic name: Butan-2-ol [1]	2
38(c)	Structure: CH ₃ -CH ₂ -CO-CH ₃ [1] Systematic name: Butanone [1] Test: Add 2,4-DNPH to observe an orange precipitate (and negative Tollens' test confirms ketone) [1]	3
39(a)	2-HOC ₆ H ₄ COOH + (CH ₃ CO) ₂ O → 2-CH ₃ COOC ₆ H ₄ COOH + CH ₃ COOH [2]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
39(b)	Catalyst: Concentrated phosphoric acid (H ₃ PO ₄) or conc. H ₂ SO ₄ [1] Water bath ensures gentle, even heating without thermal decomposition or ignition [1]	2
39(c)	Pour reaction mixture onto ice water; aspirin precipitates as a crude solid [1] Filter under reduced pressure (Buchner funnel) and recrystallise from minimum hot water/ethanol [1]	2
40(a)	(i) % Yield = (actual mass obtained / theoretical maximum mass) x 100% [1] (ii) % Atom Economy = (Mr of desired product / sum of Mr of all reactants) x 100% [1]	2
40(b)	Sum of Mr of all reactants = 74.1 + 102.9 + 98.1 = 275.1 [1] Mr of desired product (1-bromobutane) = 137.0 [1] Atom economy = (137.0 / 275.1) x 100 = 49.8% [1]	3
40(c)	Moles of butan-1-ol = 14.8 / 74.1 = 0.1997 mol [1] Theoretical yield of 1-bromobutane = 0.1997 x 137.0 = 27.36 g [1] Percentage yield = (19.2 / 27.36) x 100 = 70.2% [1]	3
40(d)	Even if all reactant molecules are converted completely to products (100% yield), a low atom economy means significant mass is converted into unwanted, stoichiometric by-products (e.g. NaHSO ₄ , salts) [1] These unwanted by-products represent chemical waste requiring disposal or energy-intensive separation [1]	2
41(a)	100% atom economy [1] It is an addition reaction; all atoms in the reactants (ethene and water) are combined into the single desired ethanol product with zero by-products [1]	2
41(b)	Total reactant mass = 108.9 + 40.0 = 148.9 [1] % Atom economy = (46.0 / 148.9) x 100 = 30.9% [1]	2
41(c)	Method 2 is substantially greener [1] It has 100% atom economy (compared to only 30.9% for Method 1), generating zero toxic waste (no sodium bromide waste), conserving raw materials [1]	2
42(a)	Chemical products should be designed so that at the end of their function they break down into innocuous environmental degradation products and do not persist [1]	1
42(b)	Catalysts are regenerated and used in small quantities, speeding up reactions and lowering activation energy [1] Stoichiometric reagents are consumed in molar quantities and produce large masses of by-product chemical waste [1]	2
42(c)	Water and CO ₂ are non-flammable, non-toxic, and readily available [1] Dichloromethane and benzene are volatile organic compounds (VOCs) that are toxic, carcinogenic, deplete the ozone layer, or present severe disposal hazards [1]	2
43(a)	Overall yield = 0.750 x 0.800 = 0.600 = 60.0% [2]	2
43(b)	Moles of propanoic acid = 11.1 / 74.1 = 0.1498 mol [1] Moles of bromoethane needed = 0.1498 / 0.600 = 0.2497 mol [1] Mass of bromoethane = 0.2497 x 108.9 = 27.2 g [1]	3
44(a)	A (Polymerisation of ethene to poly(ethene)) [1]	1
44(b)	100% [1]	1
45(a)	Fewer synthetic steps (3 steps vs 6 steps), dramatically reducing solvent usage and increasing overall yield [1] Substantially higher atom economy (77% vs 40%), using recyclable catalysts (HF, Raney nickel, palladium) rather than stoichiometric reagents like AlCl ₃ [1]	2
45(b)	Less raw chemical materials are consumed per kilogram of drug produced [1] Dramatically reduces the volume of hazardous chemical waste and by-products that must be incinerated or neutralised [1]	2
46(a)	Total reactant Mr = 60.0 + 46.0 = 106.0 [1] Atom economy = (88.0 / 106.0) x 100 = 83.0% [1]	2
46(b)	Total reactant Mr = 78.5 + 46.0 = 124.5 [1] Atom economy = (88.0 / 124.5) x 100 = 70.7% [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
46(c)	The carboxylic acid route has a higher theoretical atom economy (83.0% vs 70.7%) and produces benign water rather than corrosive HCl [1] However, the acyl chloride route goes to 100% completion irreversibly, whereas the carboxylic acid route is limited by dynamic equilibrium [1]	2
47(a)	A homogeneous catalyst is in the same physical state / phase as the reactants (e.g. liquid acid in liquid reactants) [1] A heterogeneous catalyst is in a different physical state / phase from the reactants (e.g. solid catalyst with liquid or gaseous reactants) [1]	2
47(b)	Solid Al ₂ O ₃ is easily separated from the product gas stream by simple phase filtration without neutralisation waste [1] Avoids corrosive, hazardous liquid H ₂ SO ₄ which produces acid waste and toxic SO ₂ by-product fumes [1]	2
48(a)	C (100%) [1]	1
48(b)	Two reactant molecules combine to form a single product molecule with no atoms eliminated or lost as waste [1]	1
49(a)	E-factor = 1,250 kg / 25.0 kg = 50.0 [2]	2
49(b)	Lower E-factor [1] A lower E-factor means less chemical waste is generated per unit mass of desired product produced, indicating higher material efficiency [1]	2
49(c)	Organic solvents used in reaction media, extractions, and chromatography columns [1] Inorganic salts and reagents from quenching, neutralisation, and washing stages [1]	2
50(a)	Reagents/conditions: Steam (H ₂ O(g)) and concentrated H ₃ PO ₄ catalyst at 300 °C, 60 atm [2]	2
50(b)	Acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄), heat under reflux [1] CH ₃ CH(OH)CH ₂ CH ₃ + [O] -> CH ₃ COCH ₂ CH ₃ + H ₂ O [1]	2
50(c)	Reagents: HCN with NaCN (or KCN) catalyst at room temperature, pH ~ 8 [2] Mechanism: Nucleophilic addition [1]	3
50(d)	Reagents/conditions: Dilute hydrochloric acid (dilute HCl(aq)), heat under reflux [1] Planar carbonyl group of butanone is attacked with equal probability from either face by the :CN nucleophile [1] Produces an equimolar (50:50) racemic mixture of both optical enantiomers, cancelling optical rotation [1]	3